

Applying Data Science Techniques to Analyze Bike-Share Usage

Purpose of the study

This study examined daily bike-share usage to better understand what factors drive ridership and how accurately statistical models can predict demand. The dataset includes two years of observations (2011-2012) with variables describing weather conditions, seasonality, and daily bike counts. The goal was to identify key environmental and seasonal conditions that influence bike usage and evaluate whether data-driven models can help forecast future ridership.

Data from 2011 were used to train models and identify patterns in usage related to weather, seasonality, and calendar effects. The models were then tested using 2012 data to assess how well they perform when predicting real-world demand in a new time period.

Findings

The analysis showed that weather and seasonal conditions are the strongest drivers of bike-share usage. Warmer temperatures are associated with higher ridership, while high humidity, strong winds, and poor weather conditions tend to reduce usage. Ridership is also significantly higher during the summer and fall months compared to winter.

Several statistical modeling approaches were tested to predict daily bike counts. While these models performed reasonably well when trained on the 2011 data, their accuracy declined when applied to the 2012 data. Predictions tended to underestimate higher ridership days, indicating that overall demand increased in 2012 and that models trained on earlier data did not fully capture this change. The decline in performance suggests that the underlying demand structure shifted between years, rather than remaining stable.

Recommendation

These results highlight two important insights for bike-share system planning. First, weather and seasonal patterns strongly influence daily demand, which has practical implications for system operations. For example, operators may expect higher ridership on warm summer days and can prepare by ensuring more bikes are available at popular stations, scheduling additional maintenance staff, and planning for higher usage. In contrast, colder or stormy days may require fewer bikes in circulation and less operational capacity.

Second, predictive models should be updated regularly with recent data, since demand patterns can change as the bike-share system grows. The drop in predictive performance from 2011 to 2012 shows that models trained on a single year may not capture evolving usage trends.

To improve future predictions, organizations should:

- Incorporate multiple years of data when training models
- Include variables that capture system growth or long-term trends
- Regularly retrain models to reflect current usage patterns

By combining updated data with consistent model evaluation, bike-share systems can improve forecasting accuracy and make more informed decisions about resource allocation, staffing, and system expansion.